

Milestone 3

Critical Evaluation and Transfer

I. GENERAL INFORMATION

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Topic:	Dynamic Priority Exchange Server
Seed reference:	Hard Real-Time Computing Systems - Predictable Scheduling Algorithms and Applications Giorgio Buttazzo

II. PROBLEM STATEMENT AND MOTIVATION

Since the first two milestones were completed, it is now known how the Dynamic Priority Exchange Server works and how it compares with the scheduling approaches in *Milestone 2*.

The goal of this milestone is to further analyze this scheduling algorithm and identify the strengths, weaknesses, assumptions, risks, and applicability of a technical contribution.

The focus is not to ultimately identify if this approach is the most optimal one, but rather for which assumptions it is useful, where it may fail, and how it transfers to realistic embedded systems contexts.

III. STRENGTHS AND LIMITATIONS

As the other scheduling approaches, DPES comes with its advantages and limitations. These can be seen below.

Strengths

- Improves responsiveness of aperiodic tasks.
- Preserves EDF schedulability guarantees.
- Overall better processor utilization than simpler approaches.

Limitations

- Increased scheduler complexity
- Additional runtime overhead
- Assumes utilization under EDF Scheduling
- Assumes task parameters are known

If server parameters are incorrect, this may lead to under-utilization of available processor capacity or poor responsiveness

IV. ASSUMPTIONS AND LIMITATIONS

Assumptions

- Tasks are executed on a single processor system
- Periodic tasks are hard real-time tasks
- Periodic tasks are schedulable under EDF
- Deadlines are predictable and deterministic
- The scheduler can dynamically update priorities during runtime

These are the same assumptions that were mentioned in Milestone 1, which are valid for this milestone as well.

V. RUNTIME, MEMORY AND SCALABILITY IMPLICATIONS

Runtime

- Additional scheduling overhead compared to standard EDF.
- More processing is required to manage server capacities and priorities.

Memory

- Requires extra memory to store server-related data.
- Memory usage increases with the number of tasks.

Scalability

- Should work well for small and medium-sized real-time systems in theory.

As the number of periodic and aperiodic tasks increases, the scheduler must maintain additional deadline and capacity information, which could increase runtime overhead and implementation complexity.

VI. DPES APPLICATION EXAMPLES

Suitable Applications

- Automotive control systems
- Industrial automation
- Robotics
- Avionics
- Systems with many periodic tasks and occasional asynchronous events.

Unsuitable Applications

- Small microcontrollers with extremely limited resources.
- Systems where simple fixed-priority scheduling is sufficient.

- Applications with almost no aperiodic workload.

DPES is most suitable for systems that must handle unpredictable events while maintaining strict timing guarantees for periodic tasks, and less suitable for simple or highly resource-constrained systems.

VII. EXTENSION OR ADAPTATION

An interesting topic to research would be adapting DPES to multi-core real-time systems, where preserving EDF schedulability guarantees becomes significantly more challenging due to task migration and shared resources. Until now the focus has been on single-core processors as mention in section IV and [1].

REFERENCES

- [1] G. Buttazzo, *Hard Real-Time Computing Systems: Predictable Scheduling Algorithms and Applications*, 3rd ed. Springer, 2011.